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Unit 1: Numerical methods

A numerical method is an approximate computer method for solving a mathematical problem which often has no analytical solution.

Analytical Solutions

- Many problems have solutions that are obvious once the problem has been defined.
- A set of logical steps that we can follow to calculate an exact outcome.
- In linear algebra, there are a suite of methods that you can use to factorize a matrix, depending on if the properties of your matrix are square, rectangular, contain real or imaginary values, and so on.
- Some problems in applied machine learning are well defined and have an analytical solution.
- For example, the method for transforming a categorical variable into a one hot encoding is simple, repeatable and always the same methodology regardless of the number of integer values in the set.
- Unfortunately, most of the problems that we care about solving in machine learning do not have analytical solutions.

Numerical Solutions

- There are many problems that we are interested in that do not have exact solutions.
- We have to make guesses at solutions and test them to see how good the solution is. This involves framing the problem and using trial and error across a set of candidate solutions.
- In essence, the process of finding a numerical solution can be described as a search

These types of solutions have some interesting properties:

- We often easily can tell a good solution from a bad solution.
- We often don't objectively know what a "good" solution looks like; we can only compare the goodness between candidate solutions that we have tested.
- We are often satisfied with an approximate or "good enough" solution rather than the single best solution.
- Often the problems that we are trying to solve with numerical solutions are challenging, where any "good enough" solution would be useful. It also highlights that there are many solutions to a problem and even that many of them may be good enough to be usable.
- Most of the problems that we are interested in solving in applied machine learning require a numerical solution.

Analytical vs Numerical Solutions

- An analytical solution involves framing the problem in a well-understood form and calculating the exact solution.
- A numerical solution means making guesses at the solution and testing whether the problem is solved well enough to stop.

- An example is the square root that can be solved both ways.
- We prefer the analytical method in general because it is faster and because the solution is exact. Nevertheless, sometimes we must resort to a numerical method due to limitations of time or hardware capacity.
- A good example is in finding the coefficients in a linear regression equation that can be calculated analytically (e.g. using linear algebra), but can be solved numerically when we cannot fit all the data into the memory of a single computer in order to perform the analytical calculation (e.g. via gradient descent).
- Sometimes, the analytical solution is unknown and all we have to work with is the numerical approach.

Numerical Solutions in Machine Learning

- Applied machine learning is a numerical discipline.
- The core of a given machine learning model is an optimization problem, which is really a search for a set of terms with unknown values needed to fill an equation. Each algorithm has a different “equation” and “terms”, using this terminology.
- The equation is easy to calculate in order to make a prediction for a given set of terms, but we don’t know the terms to use in order to get a “good” or even “best” set of predictions on a given set of data.
- This is the numerical optimization problem that we always seek to solve.
- It’s numerical, because we are trying to solve the optimization problem with noisy, incomplete, and error-prone limited samples of observations from our domain. The model is trying hard to interpret the data and create a map between the inputs and the outputs of these observations.

Types of Errors

Numerically computed solutions are subject to certain errors. Mainly there are three types of errors. They are inherent errors, truncation errors and errors due to rounding.

1. **Inherent errors** or experimental errors arise due to the assumptions made in the mathematical modelling of problem. It can also arise when the data is obtained from certain physical measurements of the parameters of the problem. i.e., errors arising from measurements.
2. **Truncation errors** are those errors corresponding to the fact that a finite (or infinite) sequence of computational steps necessary to produce an exact result is “truncated” prematurely after a certain number of steps.
3. **Round off errors** are errors arising from the process of rounding off during computation. These are also called chopping, i.e. discarding all decimals from some decimals on.

Errors in Numerical Procedures

The relationship between the exact, or true, result and the approximation can be formulated as

True value = approximation + error

We get,

$$E_t = | \text{True value} - \text{approximation} |$$

where E_t is used to designate the exact value of the absolute error. The subscript t is included to designate that this is the “absolute true” error.

- A shortcoming of this definition is that it takes no account of the order of magnitude of the value under examination.

- Absolute True relative error = $\left| \frac{\text{true error}}{\text{true value}} \right|$

- Absolute percent relative error $\varepsilon_T = \left| \frac{\text{true error}}{\text{true value}} \right| \times 100$

- However, in machine learning applications, we will obviously not know the true answer beforehand. For these situations, an alternative is to normalize the error using the best available estimate of the true value, that is, to the approximation itself,

$$\varepsilon_a = \frac{\text{approximate error}}{\text{approximation}} 100\%$$

- Certain numerical methods use an iterative approach to compute answers. In such an approach, a present approximation is made on the basis of a previous approximation. This process is performed repeatedly, or iteratively, to successively compute better and better approximations. For such cases, the error is often estimated as the difference between previous and current approximations. Thus, percent relative error is determined according to

- $\varepsilon_a = \left| \frac{\text{current approximation} - \text{previous approximation}}{\text{current approximation}} \right| \times 100$

- When performing computations, we may not be concerned with the sign of the error, but we are interested in whether the percent absolute value is lower than a prespecified percent tolerance ε .

$$|\varepsilon_a| < \varepsilon$$

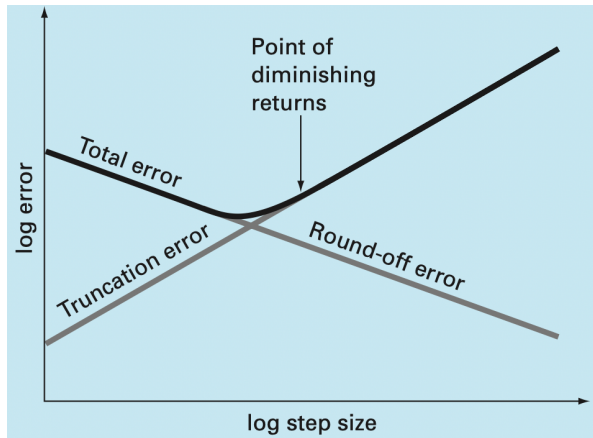
We say that the estimate is correct to n decimal digits if:

$$|\text{Error}| \leq 10^{-n}$$

We say that the estimate is correct to n decimal digits **rounded** if:

$$|\text{Error}| \leq \frac{1}{2} \times 10^{-n}$$

- The total numerical error is the summation of the truncation and round-off errors. In general, the only way to minimize round-off errors is to increase the number of significant figures of the computer. Further, we have noted that round-off error will increase due to subtractive cancellation or due to an increase in the number of computations in an analysis. In contrast, the truncation error can be reduced by decreasing the step size. Because a decrease in step size can lead to subtractive cancellation or to an increase in computations, the truncation errors are decreased as the round-off errors are increased.



Root Finding Methods

- An important problem in applied mathematics is to "solve $f(x) = 0$ " where $f(x)$ is a function of x .
- The values of x that make $f(x) = 0$ are called the roots of the equation. They are also called the zeros of $f(x)$.
- The root finding methods are divided into two categories: bracketing and open methods.
- The bracketing methods require the limits between which the root lies
- Bisection and False position methods are examples of the bracketing methods.
- Open methods require the initial estimation of the solution. Among the open methods, the Newton-Raphson and Secant is most commonly used.
- The most popular method for solving a non-linear equation is the Newton-Raphson method and this method has a high rate of convergence to a solution.

Bisection method

Since the method is based on finding the root between two points, the method falls under the category of bracketing methods.

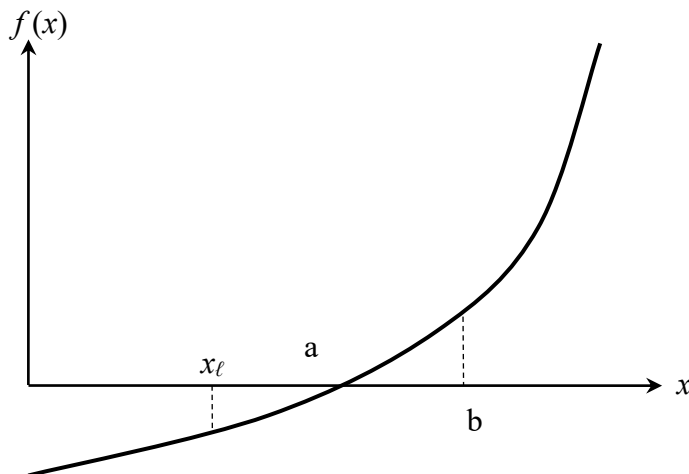


Figure 1 At least one root exists between the two points if the function is real, continuous, and changes sign.

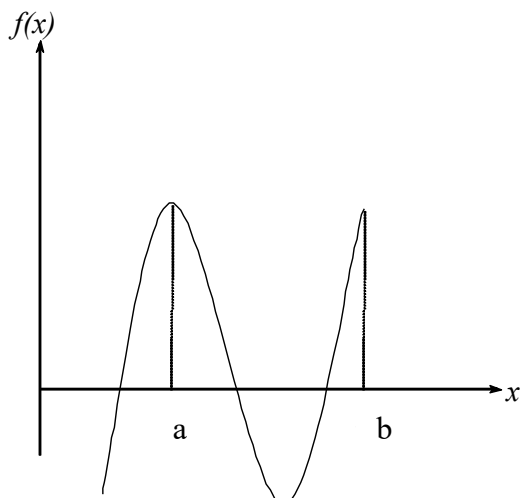


Figure 2 If the function $f(x)$ does not change sign between the two points, roots of the equation $f(x) = 0$ may still exist between the two points.

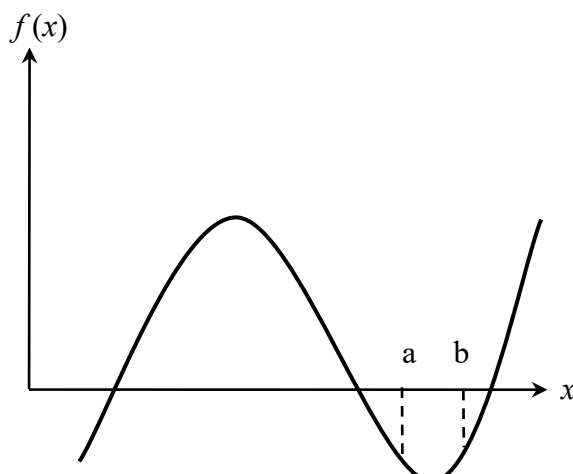
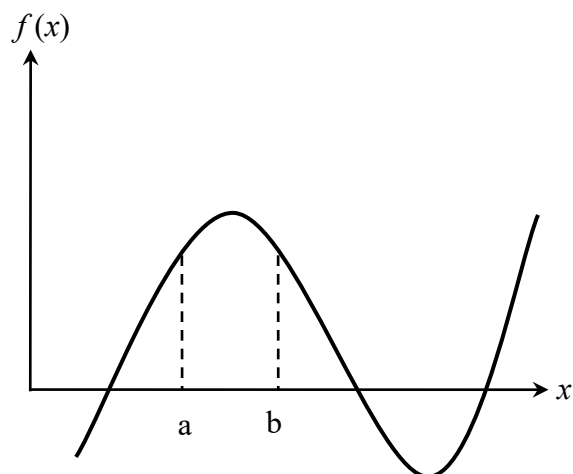


Figure 3 If the function $f(x)$ does not change sign between two points, there may not be any roots for the equation $f(x) = 0$ between the two points.

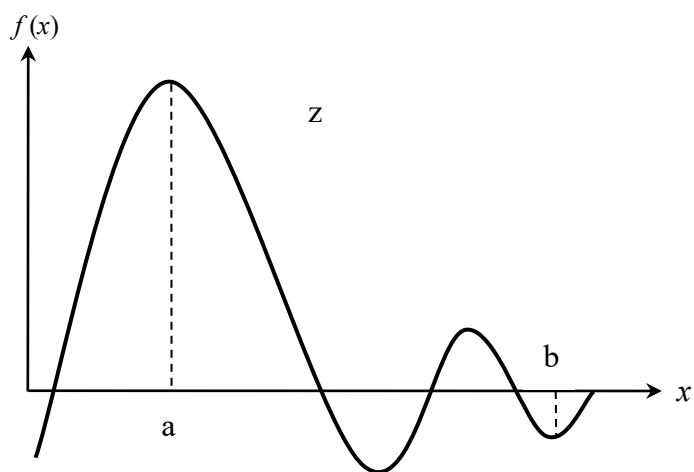


Figure 4 If the function $f(x)$ changes sign between the two points, more than one root for the equation $f(x) = 0$ may exist between the two points.

Since the root is bracketed between two points, a and b , one can find the mid-point, c between a and b . This gives us two new intervals

1. a and c , and
2. c and b .

Is the root now between a and c or between c and b ? Well, one can find the sign of $f(a).f(c)$, and if $f(a).f(c) < 0$ then the new bracket is between a and c , otherwise, it is between c and b . So, you can see that you are literally halving the interval. As one repeats this process, the width of the interval becomes smaller and smaller, and you can zero in to the root of the equation $f(x) = 0$.

The first approximation to the root is $c_1 = \frac{(a+b)}{2}$.

If $f(c_1) = 0$, then c_1 is a root of $f(x) = 0$, otherwise, the root lies in (a, c_1) or (c_1, b) according to $f(c_1)$ is (+)ve or (-)ve.

Then we bisect the interval as before and continue the process until the root is found to the desired accuracy. In the adjoining figure, $f(c_1)$ is (-)ve so that the root lies between b and c_1 .

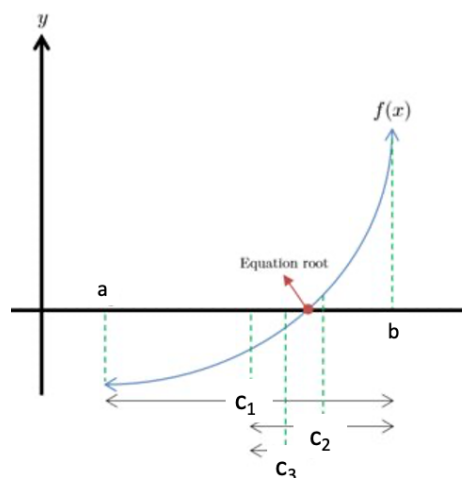
$$f(c_1) \cdot f(b) < 0$$

The second approximation to the root is $c_2 = \frac{(c_1+b)}{2}$.

$f(c_2)$ is (+)ve the root lies between c_1 and c_2 .

$$f(c_1) \cdot f(c_2) < 0$$

Similarly, the third approximation to the root is c_3 and so on. Then $c_3 = \frac{(c_1+c_2)}{2}$.



Algorithm for the bisection method

The steps to apply the bisection method to find the root of the equation $f(x) = 0$ are

1. Choose a and b as two guesses for the root such that $f(a).f(b) < 0$, or in other words, $f(x)$ changes sign between a and b . Let $k=1$
2. Estimate the root, c , of the equation $f(x) = 0$ as the mid-point between a and b as

$$c_k = \frac{a+b}{2}$$
3. Now check the following
 - a) If $f(a).f(c) < 0$, then the root lies between a and c ; then $b = c$.
 - b) If $f(a).f(c) > 0$, then the root lies between c and b ; then $a = c$

- c) If $f(c) = 0$; then the root is c . Stop the algorithm if this is true.
- Find the new estimate of the root $c_k = \frac{a+b}{2}$
 - Compare the absolute approximate error $|c_k - c_{k-1}|$ or absolute relative approximate error $\frac{|c_k - c_{k-1}|}{|c_k|}$ with the pre-specified relative error tolerance ϵ .
where
 c_k = estimated root from present iteration
 c_{k-1} = estimated root from previous iteration
 If $|c_k - c_{k-1}| < \epsilon$ or $\frac{|c_k - c_{k-1}|}{|c_k|} < \epsilon$ then exit
 Else
 $k = k + 1$
 go to Step 3
 Note one should also check whether the number of iterations is more than the maximum number of iterations allowed.

Advantages of bisection method

- The bisection method is always convergent. Since the method brackets the root, the method is guaranteed to converge.
- As iterations are conducted, the interval gets halved. So one can guarantee the error in the solution of the equation.

Drawbacks of bisection method

- The convergence of the bisection method is slow as it is simply based on halving the interval.
- If one of the initial guesses is closer to the root, it will take larger number of iterations to reach the root.
- If a function $f(x)$ is such that it just touches the x -axis (Figure 5) such as

$f(x) = x^2 = 0$ it will be unable to find a , and b such that $f(a).f(b) < 0$

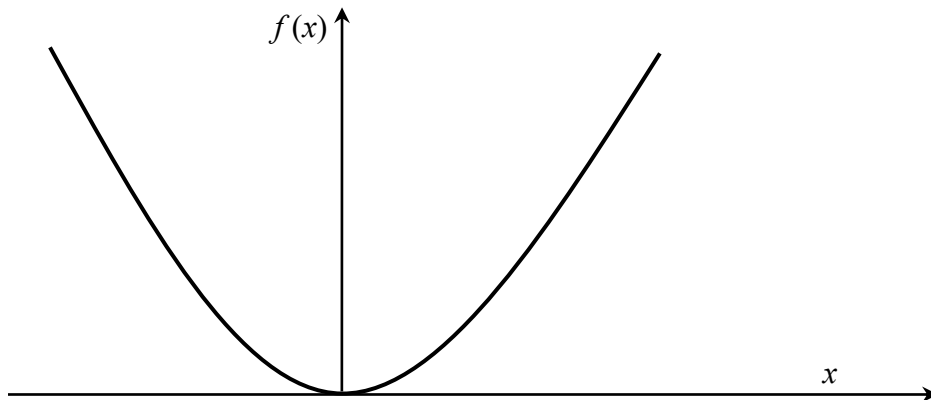


Figure 5 The equation $f(x) = x^2 = 0$ has a single root at $x = 0$ that cannot be bracketed.

- For example $f(x) = \frac{1}{x}$ where $a = -2$ and $b = -3$, are valid initial guesses which satisfy $f(a).f(b) < 0$

However, the function is not continuous and the theorem that a root exists is also not applicable.

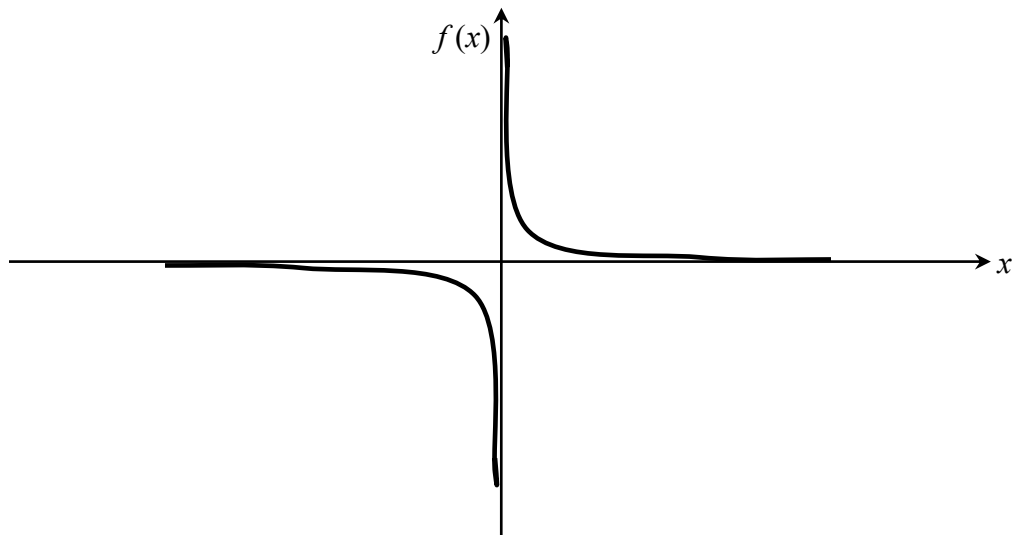


Figure 6 The equation $f(x) = \frac{1}{x} = 0$ has no root but changes sign.

Example

$$x^4 - 3x^2 + x - 10 = 0$$

Find an interval of unit length which contains this root.

Solution

$$f(0) = 0 - 0 + 0 - 10 = -10$$

$$f(1) = 1 - 3 + 1 - 10 = -11$$

$$f(2) = 16 - 12 + 2 - 10 = -4$$

$$f(3) = 81 - 27 + 3 - 10 = 47$$

An interval of unit length which contains the root of

$x^4 - 3x^2 + x - 10 = 0$ is (2,3) as $f(2)$ and $f(3)$ are of opposite sign

Example

Find the real root of the equation $x^3 - 4x - 9 = 0$ by Bisection method correct. Take $a = 2.706$, $b = 2.707$, $\epsilon = 0.0001$

Solution

$f(x) = x^3 - 4x - 9$ is given

Iteration 1:

Here, we take $k = 1$, $a = 2.706$, $b = 2.707$

$$f(a) = f(2.706) = (2.706)^3 - 4(2.706) - 9 = -0.009488 \text{ i.e., } (-)\text{ve}$$

and

$$f(b) = f(2.707) = (2.707)^3 - 4(2.707) - 9 = 0.008487 \text{ i.e., } (+)\text{ve}$$

As $f(2.706)$ and $f(2.707)$ are of opposite signs, the root lies between 2.706 and 2.707.

First approximation to the root is

$$c = \frac{a+b}{2} = \frac{(2.706+2.707)}{2} = 2.7065$$

$$f(c) = f(2.7065) = (2.7065)^3 - 4(2.7065) - 9 = -0.0005025$$

For the first iteration error = $|c_k - c_{k-1}|$ does not exist

Putting the values in a tabular form we get

k	a	f(a)	b	f(b)	$c = (a+b)/2$	f(c)	$ c_k - c_{k-1} $
1	2.706	-0.009488	2.707	0.008487	2.7065	-0.0005025	-

Iteration 2:

$$k = 2$$

Now, $f(c) = -0.0005025$ i.e., (-)ve and $f(b) = 0.008487$ i.e., (+)ve

Hence, the root lies between 2.7065 and 2.707.

So we take $a = 2.7065$ and $b = 2.707$

Second approximation to the root is

$$c = \frac{a+b}{2} = \frac{(2.7065+2.707)}{2} = 2.70675$$

$$f(c) = f(2.70675) = (2.70675)^3 - 4(2.70675) - 9 = 0.003992$$

$$\text{Error} = |c_k - c_{k-1}| = |2.70675 - 2.7065| = 0.00025$$

Here, $\epsilon > \text{Error}$, therefore we perform third iteration

Putting the values in a tabular form we get

k	a	f(a)	b	f(b)	$c = (a+b)/2$	f(c)	$ c_k - c_{k-1} $
1	2.706	-0.009488	2.707	0.008487	2.7065	-0.0005025	-
2	2.7065	-0.0005025	2.707	0.008487	2.70675	0.003992	0.00025

Iteration 3:

$$k = 3$$

Now $f(c) = 0.003992$ i.e., (+)ve and $f(a) = -0.0005025$ i.e., (-)ve

Hence, the root lies between 2.7065 and 2.70675.

So we take $a = 2.7065$ and $b = 2.70675$

Third approximation to the root is

$$c = \frac{a+b}{2} = \frac{(2.7065+2.70675)}{2} = 2.706625$$

$$\text{Now } f(c) = 0.001744$$

$$f(c) = f(2.706625) = (2.706625)^3 - 4(2.706625) - 9 = 0.001744$$

$$\text{Error} = |c_k - c_{k-1}| = |2.706625 - 2.70675| = 0.000125$$

Here, $\epsilon > \text{Error}$, therefore we perform fourth iteration

Putting the values in a tabular form we get

k	a	f(a)	b	f(b)	$c = (a+b)/2$	f(c)	$ c_k - c_{k-1} $
1	2.706	-0.009488	2.707	0.008487	2.7065	-0.0005025	-
2	2.7065	-0.0005025	2.707	0.008487	2.70675	0.003992	0.00025
3	2.7065	-0.0005025	2.70675	0.003992	2.706625	0.001744	0.000125

Iteration 4:

$$k = 4$$

Now $f(c) = 0.001744$ i.e., (+)ve and $f(a) = -0.0005025$ i.e., (-)ve

Hence, the root lies between 2.7065 and 2.706625.

So we take $a = 2.7065$ and $b = 2.706625$

Fourth approximation to the root is

$$c = \frac{a+b}{2} = \frac{(2.7065+2.7406625)}{2} = 2.7065625$$

$$|c_k - c_{k-1}| = 0.0000625 < \epsilon$$

Hence, the root is 2.7065625, correct to three decimal places.

Putting the values in a tabular form we get

k	a	f(a)	b	f(b)	$c = (a+b)/2$	f(c)	$ c_k - c_{k-1} $
1	2.706	-0.009488	2.707	0.008487	2.7065	-0.0005025	-
2	2.7065	-0.0005025	2.707	0.008487	2.70675	0.003992	0.00025
3	2.7065	-0.0005025	2.70675	0.003992	2.706625	0.001744	0.000125
4	2.7065	-0.0005025	2.706625	0.001744	2.7065625		0.0000625

Example

The quadratic $(x - 0.3)(x - 0.5)$ obviously has zeros at 0.3 and 0.5.

- Why is the interval $[0.1, 0.6]$ not a satisfactory starting interval for bisection?
- What are good starting intervals for each root?
- If you start with $[0, 0.491]$ which root is reached with bisection? Which root is reached from $[0.31, 1.0]$?

Solution

- Both a and b have same sign
- $[0.2, 0.4]$ for 0.3 and $[0.4, 0.6]$ for 0.5
- 0.3, 0.5

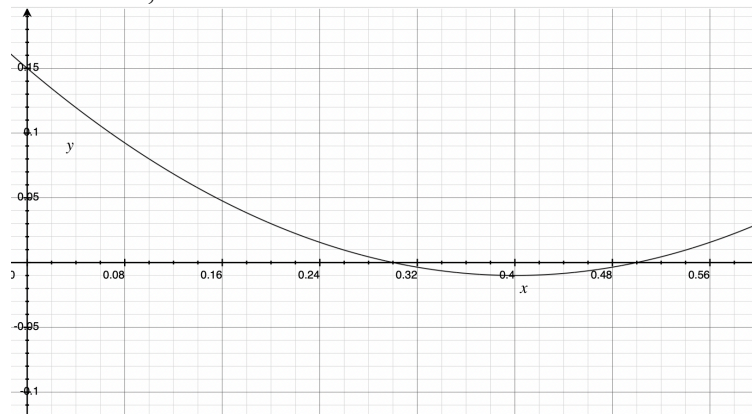


Figure 7 Graph of the function

Example

Find the real root of the equation $x^3 - x - 1$ by Bisection method correct to one decimal places.

Take $a = 1.25$, $b = 1.5$,

Solution

Accuracy of one decimal place is desired, hence $\epsilon = 0.01$

Summarizing the solution in a tabular form below:

a	f(a)	b	f(b)	$c = (a+b)/2$	f(c)	$ c_k - c_{k-1} $
1.25	-0.2969	1.5	0.875	1.375	0.2246	-
1.25	-0.2969	1.375	0.2246	1.3125	-0.0515	0.0625
1.3125	-0.0515	1.375	0.2246	1.3438	0.0826	0.0313
1.3125	-0.0515	1.3438	0.0826	1.3281	0.0146	0.0157
1.3125	-0.0515	1.3281	0.0146	1.3203		0.0078

False-Position Method or Regula Falsi Method

The false-position method takes advantage of this observation mathematically by drawing a straight line from the function value at a to the function value at b, and estimates the root as where it crosses the x-axis.

The point, where it cuts X axis is, the new estimate of the root c. Mathematically, estimate of formula for c is as follows:

$$c = \frac{af(b) - bf(a)}{f(b) - f(a)}$$

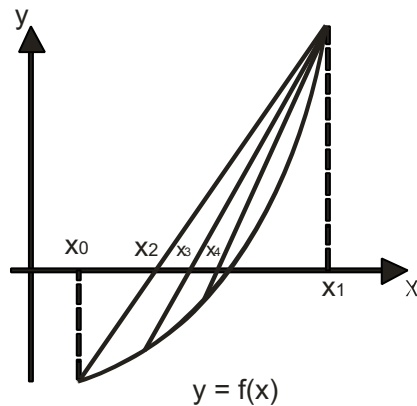


Figure 8 False-Position Method

False-Position Algorithm

The steps to apply the false-position method to find the root of the equation $f(x) = 0$ are as follows.

1. Choose a and b as two guesses for the root such that $f(a).f(b) < 0$, or in other words, $f(x)$ changes sign between a and b. Let $k = 1$
2. Estimate the root, c, of the equation $f(x) = 0$ as the mid-point between a and b as

$$c_k = \frac{af(b) - bf(a)}{f(b) - f(a)}$$
3. Now check the following
 - d) If $f(a).f(c) < 0$, then the root lies between a and c; then $b = c$.
 - e) If $f(a).f(c) > 0$, then the root lies between c and b; then $a = c$
 - f) If $f(c) = 0$; then the root is c. Stop the algorithm if this is true.
4. Find the new estimate of the root

$$c_k = \frac{af(b) - bf(a)}{f(b) - f(a)}$$

5. Compare the absolute approximate error $|c_k - c_{k-1}|$ or absolute relative approximate error $\frac{|c_k - c_{k-1}|}{|c_k|}$ with the pre-specified relative error tolerance ϵ .

where

c_k = estimated root from present iteration

c_{k-1} = estimated root from previous iteration

If $|c_k - c_{k-1}| < \epsilon$ or $\frac{|c_k - c_{k-1}|}{|c_k|} < \epsilon$ then exit

Else

$K = k + 1$

go to Step 3

Note one should also check whether the number of iterations is more than the maximum number of iterations allowed. If so, one needs to terminate the algorithm and notify the user about it.

Different Stopping Criterion

The sequence of c_k 's approaches the root. That is why, in case of the Bisection section algorithm, iterative process was stopped, when $|c_k - c_{k-1}| < \epsilon$

It is called x – tolerance criterion (X-TOL). The recent most c_k is the estimate of the root. There is another stopping criterion called function tolerance (F-TOL). At root, function value is zero, so if estimate is quite near the root then function value would be small. Hence, many times, one would like to stop when $|f(c_k)| < \delta$ where δ is some small positive constant like ϵ .

If slope of the curve is small near the root, curve is almost horizontal, and then function tolerance may not be appropriate stopping criterion, because curve is rising slowly, function values in neighbourhood of the root are going to be small, so even if estimate c is not sufficiently near the root, one may stop.

On the other hand, if it is expected that slope is high near the root or sequence of approximations c_k 's may converge to the root, like in case of almost vertical graph, then function tolerance ensures that estimate is good approximation to the root.

In simple words, for $f(x)$, if $f'(x)$ is high near the root, FTOL would be better to use as stopping criterion

Many times instead of absolute error $|c_k - c_{k-1}|$, bound on Relative error $\frac{|c_k - c_{k-1}|}{|c_k|} < \epsilon$ is

used. This needs to be applied when looking answer correct to certain number of significant digits. For example, the root could be like 1.2749×10^{-12}

So here, if we go for $|c_k - c_{k-1}| < 10^{-5}$ we shall get answer zero, but if we go for $\frac{|c_k - c_{k-1}|}{|c_k|} < 10^{-5}$, we would get root correct to 5 significant digits. It root is required to be correct to

N significant digits, then one should apply

$$\frac{|c_k - c_{k-1}|}{|c_k|} < 10^{-N}$$

Advantages:

1. It is faster than Bisection Method.
2. It is also simple.
3. It guarantees convergence.
4. Only one function evaluation per iteration is required.

Disadvantages:

1. Not self starting. One needs two initial guesses a and b such that $f(a) \cdot f(b) < 0$.
2. Though, faster than Bisection Method, still regarded as slow.
3. In rare cases, it may become slower than Bisection Method(Figure 9).

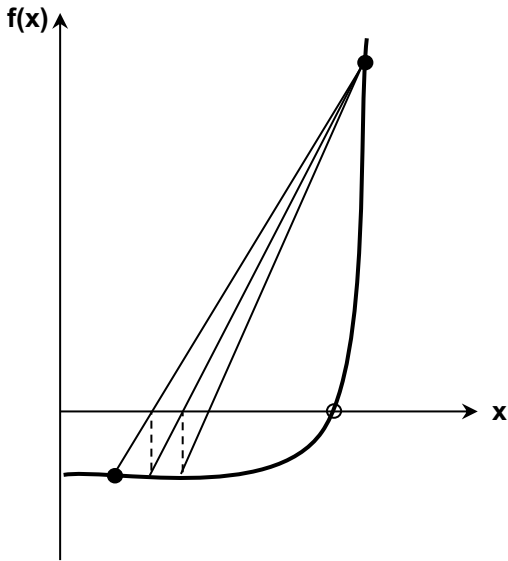


Figure 9. Slow convergence of the false-position method for $f(x) = x^{10} - 1 = 0$

Example

Find the real root of the equation $x^3 - 4x - 9 = 0$ by False position method correct.

Take $a = 2$, $b = 3$, $\epsilon = 0.01$

Solution

Iteration 1:

$k = 1$, $a = 2$, $b = 3$,

$f(a) = f(2) = (2)^3 - 4(2) - 9 = -9$ i.e., (-)ve

and

$f(b) = f(3) = (3)^3 - 4(3) - 9 = 6$ i.e., (+)ve

As $f(2)$ and $f(3)$ are of opposite signs, the root lies between 2 and 3.

First approximation to the root is

$$c = \frac{af(b) - bf(a)}{f(b) - f(a)} = \frac{(2)(6) - (3)(-9)}{6 - (-9)} = 2.6$$

$$f(c) = f(2.6) = (2.6)^3 - 4(2.6) - 9 = -1.824$$

For the first iteration error $= |c_k - c_{k-1}|$ does not exist

Putting the values in a tabular form we get

k	a	f(a)	b	f(b)	c	f(c)	$ c_k - c_{k-1} $
1	2	-9	3	6	2.6	-1.824	-

Iteration 2:

$k = 2$

Now, $f(c) = -1.824$ i.e., (-)ve and $f(b) = 6$ i.e., (+)ve

Hence, the root lies between 2.6 and 3.

So we take $a = 2.6$ and $b = 3$

Second approximation to the root is

$$c = \frac{af(b) - bf(a)}{f(b) - f(a)} = \frac{(2.6)(6) - (3)(-1.824)}{6 - (-1.824)} = 2.6933$$

$$f(c) = f(2.6933) = (2.6933)^3 - 4(2.6933) - 9 = -0.2372$$

$$\text{Error} = |c_k - c_{k-1}| = |2.6933 - 2.6| = 0.0933$$

Here, $\epsilon > \text{Error}$, therefore we perform third iteration

Putting the values in a tabular form we get

k	a	f(a)	b	f(b)	c	f(c)	$ c_k - c_{k-1} $
1	2	-9	3	6	2.6	-1.824	-
2	2.6	-1.824	3	6	2.6933	-0.2372	0.0933

Iteration 3:

k = 3

Now, $f(c) = -0.2372$ i.e., (-)ve and $f(b) = 6$ i.e., (+)ve

Hence, the root lies between 2.6933 and 3.

So we take a = 2.6933 and b = 3

Third approximation to the root is

$$c = \frac{af(b) - bf(a)}{f(b) - f(a)} = \frac{(2.6933)(6) - (3)(-0.2372)}{6 - (-0.2372)} = 2.7049$$

$$f(c) = f(2.7049) = (2.7049)^3 - 4(2.7049) - 9 = -0.028$$

$$\text{Error} = |c_k - c_{k-1}| = |2.7049 - 2.6933| = 0.0166$$

Here, $\epsilon > \text{Error}$, therefore we perform third iteration

Putting the values in a tabular form we get

k	a	f(a)	b	f(b)	c	f(c)	$ c_k - c_{k-1} $
1	2	-9	3	6	2.6	-1.824	-
2	2.6	-1.824	3	6	2.6933	-0.2372	0.0933
3	2.6933	-0.2372	3	6	2.7049	-0.0289	0.0116

Iteration 4:

k = 4

Now, $f(c) = -0.028$ i.e., (-)ve and $f(b) = 6$ i.e., (+)ve

Hence, the root lies between 2.7049 and 3.

So we take a = 2.7049 and b = 3

Fourth approximation to the root is

$$c = \frac{af(b) - bf(a)}{f(b) - f(a)} = \frac{(2.7049)(6) - (3)(-0.0289)}{6 - (-0.0289)} = 2.7063$$

$$f(c) = f(2.7063) = (2.7063)^3 - 4(2.7063) - 9 = -0.0035$$

$$\text{Error} = |c_k - c_{k-1}| = |2.7063 - 2.7049| = 0.0014$$

Here, $\text{Error} < \epsilon$

Hence, the root is 2.7063, correct to one decimal places.

Putting the values in a tabular form we get

k	a	f(a)	b	f(b)	c	f(c)	$ c_k - c_{k-1} $
1	2	-9	3	6	2.6	-1.824	-
2	2.6	-1.824	3	6	2.6933	-0.2372	0.0933
3	2.6933	-0.2372	3	6	2.7049	-0.0289	0.0116
4	2.7049	-0.0289	3	6	2.7063	-0.0035	0.0014

Example

Find the real root of the equation $x^3 - x - 1$ by False position method correct to three decimal places. $\epsilon = 0.001$

a	f(a)	b	f(b)	c	f(c)	$ c_k - c_{k-1} $
1	-1	1.5	0.875	1.2667	-0.2344	-
1.2667	-0.2344	1.5	0.875	1.316	-0.037	0.0493
1.316	-0.037	1.5	0.875	1.3234	-0.0055	0.0074
1.3234	-0.0055	1.5	0.875	1.3245	-0.0008	0.0011
1.3245	-0.0008	1.5	0.875	1.3247	-0.0001	0.0002

$$\epsilon = 0.001 \quad |c_k - c_{k-1}| = 0.0078 < \epsilon$$

The approximate real root is 1.3247

Newton-Raphson Method

Methods such as the bisection method and the false position method of finding roots of a nonlinear equation $f(x)=0$ require bracketing of the root by two guesses. Such methods are called bracketing methods. These methods are always convergent since they are based on reducing the interval between the two guesses so as to zero in on the root of the equation.

In the Newton-Raphson method, the root is not bracketed. In fact, only one initial guess of the root is needed to get the iterative process started to find the root of an equation. The method hence falls in the category of open methods. Convergence in open methods is not guaranteed but if the method does converge, it does so much faster than the bracketing methods.

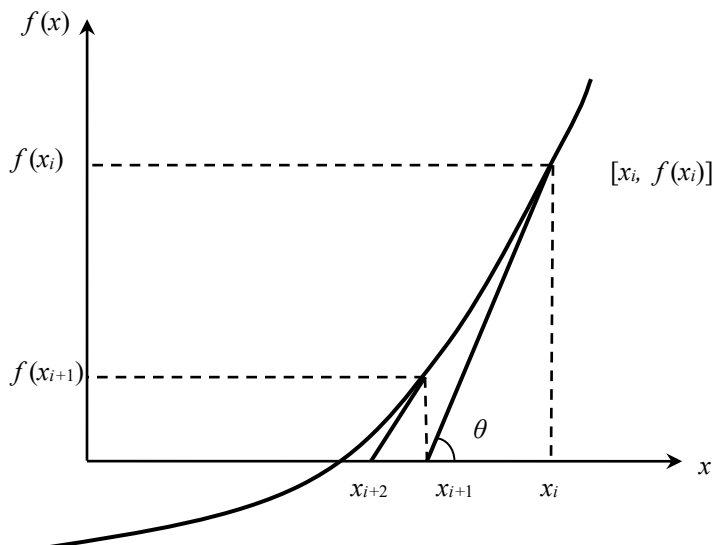


Figure 10 Newton-Raphson method.

The Newton-Raphson method is based on the principle that if the initial guess of the root of $f(x)=0$ is at x_i , then if one draws the tangent to the curve at $f(x_i)$, the point x_{i+1} where the tangent crosses the x -axis is an improved estimate of the root (Figure 10).

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

One can repeat this process until one finds the root within a desirable tolerance.

Algorithm

The steps of the Newton-Raphson method to find the root of an equation $f(x)=0$ are

1. Let $k=1$, Evaluate $f'(x)$
2. Use an initial guess of the root, x_i , to estimate the new value of the root, x_{i+1} , as

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$
3. Compare the absolute approximate error $|x_k - x_{k-1}|$ or absolute relative approximate error $\frac{|x_k - x_{k-1}|}{|x_k|}$ with the pre-specified relative error tolerance ϵ .

where

x_k = estimated root from present iteration

x_{k-1} = estimated root from previous iteration

If $|x_k - x_{k-1}| < \epsilon$ or $\frac{|x_k - x_{k-1}|}{|x_k|} < \epsilon$ then exit

Else

$k = k + 1$

go to Step 2

Note one should also check whether the number of iterations is more than the maximum number of iterations allowed. If so, one needs to terminate the algorithm and notify the user about it.

Advantages:

1. Only one initial guess is needed.
2. Very rapid convergence. It has quadratic convergence which implies, number of correct figures in the estimate is nearly doubled at each successive step.

Disadvantages:

1. If initial guess is not sufficiently near the root, it may diverge.
2. Function must be differentiable.
3. In case of multiple roots, pace of convergence slows down,
4. There can be many instances, when Newton Raphson may fail to converge,

Example

Find the real root of the equation $x^3 - 4x - 9 = 0$ by Newton Raphson method correct to two decimal places. Take $x_0 = 2.5$, $\epsilon = 0.001$

Solution

$$f'(x) = 3x^2 - 4$$

Iteration 1:

k = 1

$$f(x_i) = f(2.5) = (2.5)^3 - 4(2.5) - 9 = -3.375$$

$$f'(x_i) = f'(2.5) = 3(2.5)^2 - 4 = 14.75$$

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)} = 2.5 - \frac{-3.375}{14.75} = 2.7288$$

For the first iteration error = $|x_k - x_{k-1}|$ does not exist

Putting the values in a tabular form we get

k	x_i	$f(x_i)$	$f'(x_i)$	x_{i+1}	$ x_k - x_{k-1} $
1	2.5	-3.375	14.75	2.7288	-

Iteration 2:

k = 2

Now we take $x_i = 2.7288$

$$f(x_i) = f(2.7288) = (2.7288)^3 - 4(2.7288) - 9 = 0.4046$$

$$f'(x_i) = f'(2.7288) = 3(2.7288)^2 - 4 = 18.3393$$

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)} = 2.7288 - \frac{0.4046}{18.3393} = 2.7067$$

$$\text{Error} = |x_k - x_{k-1}| = |2.7067 - 2.7288| = 0.0221$$

Here, $\epsilon > \text{Error}$, therefore we perform third iteration

Putting the values in a tabular form we get

k	x_i	$f(x_i)$	$f'(x_i)$	x_{i+1}	$ x_k - x_{k-1} $
1	2.5	-3.375	14.75	2.7288	-
2	2.7288	0.4046	18.3393	2.7067	0.0221

Iteration 3:

k = 3

Now we take $x_i = 2.7067$

$$f(x_i) = f(2.7067) = (2.7067)^3 - 4(2.7067) - 9 = 0.004$$

$$f'(x_i) = f'(2.7067) = 3(2.7067)^2 - 4 = 17.9795$$

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)} = 2.7067 - \frac{0.004}{17.9795} = 2.7065$$

$$\text{Error} = |x_k - x_{k-1}| = |2.7065 - 2.7067| = 0.0002$$

Here, $\text{Error} < \epsilon$

Hence, the root is 2.7065, correct to two decimal places.

Putting the values in a tabular form we get

k	x_i	$f(x_i)$	$f'(x_i)$	x_{i+1}	$ x_k - x_{k-1} $
1	2.5	-3.375	14.75	2.7288	-
2	2.7288	0.4046	18.3393	2.7067	0.0221
3	2.7067	0.004	17.9795	2.7065	0.0002

Example

Find the real root of the equation $x^3 - x - 1$ by Newton Raphson method correct to three decimal places. Take $x_0 = 1.5$,

$$\epsilon = 0.001$$

$$f'(x) = 3x^2 - 1$$

k	x_i	$f(x_i)$	$f'(x_i)$	x_{i+1}	$ x_k - x_{k-1} $
1	1.5	0.875	5.75	1.3478	-
2	1.3478	0.1007	4.4499	1.3252	0.0226
3	1.3252	0.0021	4.2646	1.3247	0.0005

Example

This quadratic has two nearly equal roots:

$$P(x) \text{ is } x^2 - 4x + 3.9999$$

- Which root do you get with Newton's method starting at $x = 2.1$?
- Repeat part (a) but starting with $x = 1.9$.
- What happens with Newton's method starting from $x = 2.0$?

Solution

- Starting from $x_0 = 2.1$, convergence is to $x = 2.0108$
- Starting from $x_0 = 1.9$, convergence is to $x = 1.9892$.
- Starting from $x_0 = 2.0$ fails, $f(2.0) = \text{zero}$.

Secant Method

Secant method is an open method. That is, no more interval under consideration needs to bracket the root, though it still requires two initial guesses. Let us denote these initial guesses by x_{-1} and x_0 . The formula for generating the sequence of approximations is

$$x_{i+1} = \frac{x_{i-1}f(x_i) - x_i f(x_{i-1})}{f(x_i) - f(x_{i-1})}$$

Firstly, initial choices of x_{-1} and x_0 need not be bracketing the root. So, that is, root need not lie within the endpoints $[x_{-1}, x_0]$ or $[x_0, x_{-1}]$. That is $f(x_{-1})$ and $f(x_0)$ can be of same sign. No more, one needs to ensure that $f(x_{-1}) \cdot f(x_0) < 0$.

Though, in practice one usually chooses x_{-1}, x_0 as ones which bracket the root, there is no such compulsion. Secondly, in computation of next approximation, older approximation is discarded. Recent most two approximations are used for iteration, irrespective of function values at these end points. So graphically, straight line joining $(x_{i-1}, f(x_{i-1}))$ and $(x_i, f(x_i))$ is drawn to generate x_{i+1} , next approximation. No more, function values sign are checked to ensure that root lies between $(x_{i-1}, f(x_{i-1}))$ and $(x_i, f(x_i))$. Because secant is drawn joining $(x_{i-1}, f(x_{i-1}))$ and $(x_i, f(x_i))$, it is called Secant Method.

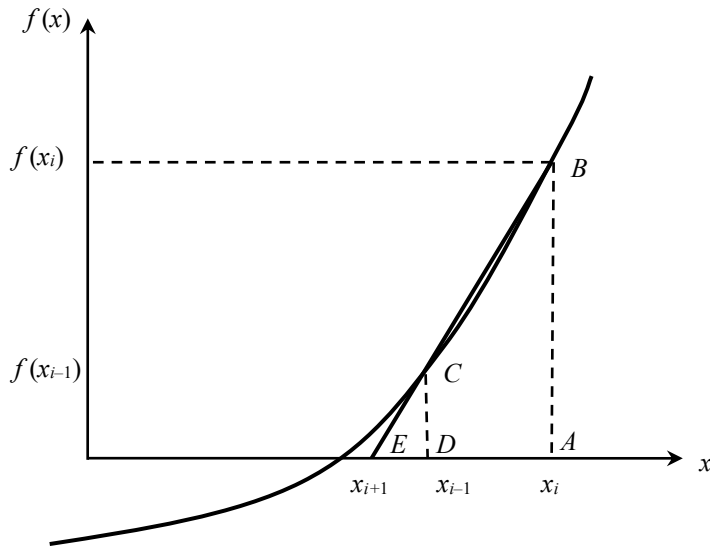


Figure 11 Geometrical representation of the secant method.

Secant Algorithm

The steps of the Secant method to find the root of an equation $f(x) = 0$ are

1. Let $k = 1$, Evaluate $f(x_{i-1})$, $f(x_i)$
2. Use an initial guess of the root, x_i , to estimate the new value of the root, x_{i+1} , as

$$x_{i+1} = \frac{x_{i-1}f(x_i) - x_i f(x_{i-1})}{f(x_i) - f(x_{i-1})}$$
3. Compare the absolute approximate error $|x_k - x_{k-1}|$ or absolute relative approximate error $\frac{|x_k - x_{k-1}|}{|x_k|}$ with the pre-specified relative error tolerance ϵ .

where

x_k = estimated root from present iteration

x_{k-1} = estimated root from previous iteration

If $|x_k - x_{k-1}| < \epsilon$ or $\frac{|x_k - x_{k-1}|}{|x_k|} < \epsilon$ then exit

Else

$K = k + 1$

go to Step 2

Note one should also check whether the number of iterations is more than the maximum number of iterations allowed. If so, one needs to terminate the algorithm and notify the user about it.

Advantages:

1. No constraint of end points of interval to converge
2. Very rapid convergence

Disadvantages:

1. If initial guess is not sufficiently near the root, it may diverge.

Difference between the secant and false-position method

In the false position method, the latest estimate of the root replaces whichever of the original values yielded a function value with the same sign as $f(x)$. The root is always bracketed by the bounds and the method will always converge.

For the secant method, x_{i+1} replaces x_i and x_i replaces x_{i-1} . As a result, the two values can sometimes lie in the same side of the root and lead to divergence. But when, the secant method converges, it usually does it at a quicker rate.

Example

Find the real root of the equation $x^3 - 4x - 9 = 0$ by Secant Method correct.

Take $x_0 = 2$, $x_1 = 3$, $\epsilon = 0.01$

Solution

Iteration 1:

$k = 1$, $x_{i-1} = 2$, $x_i = 3$,

$$f(x_{i-1}) = f(2) = (2)^3 - 4(2) - 9 = -9$$

and

$$f(x_i) = f(3) = (3)^3 - 4(3) - 9 = 6$$

First approximation to the root is

$$x_{i+1} = \frac{x_{i-1}f(x_i) - x_i f(x_{i-1})}{f(x_i) - f(x_{i-1})} = \frac{(2)(6) - (3)(-9)}{6 - (-9)} = 2.6$$

For the first iteration error = $|x_k - x_{k-1}|$ does not exist

Putting the values in a tabular form we get

k	x_{i-1}	$f(x_{i-1})$	x_i	$f(x_i)$	x_{i+1}	$f(x_{i+1})$	$ x_k - x_{k-1} $
1	2	-9	3	6	2.6	-1.824	-

Iteration 2:

$k = 2$, $x_{i-1} = 3$, $x_i = 2.6$,

$$f(x_{i-1}) = f(3) = (3)^3 - 4(3) - 9 = 6$$

and

$$f(x_i) = f(2.6) = (2.6)^3 - 4(2.6) - 9 = -1.824$$

Second approximation to the root is

$$x_{i+1} = \frac{x_{i-1}f(x_i) - x_i f(x_{i-1})}{f(x_i) - f(x_{i-1})} = \frac{(3)(-1.824) - (2.6)6}{(-1.824) - 6} = 2.6933$$

$$\text{Error} = |x_k - x_{k-1}| = |2.6933 - 2.6| = 0.0933$$

Here, $\epsilon > \text{Error}$, therefore we perform third iteration

Putting the values in a tabular form we get

k	x_{i-1}	$f(x_{i-1})$	x_i	$f(x_i)$	x_{i+1}	$f(x_{i+1})$	$ x_k - x_{k-1} $
1	2	-9	3	6	2.6	-1.824	-
2	3	6	2.6	-1.824	2.6933	-0.2372	0.0933

Iteration 3:

$k = 3$, $x_{i-1} = 2.6$, $x_i = 2.6933$,

$$f(x_{i-1}) = f(2.6) = (2.6)^3 - 4(2.6) - 9 = -1.824$$

and

$$f(x_i) = f(2.6933) = (2.6933)^3 - 4(2.6933) - 9 = -0.2372$$

Third approximation to the root is

$$x_{i+1} = \frac{x_{i-1}f(x_i) - x_i f(x_{i-1})}{f(x_i) - f(x_{i-1})} = \frac{(2.6)(-0.2372) - (2.6933)(-1.824)}{(-0.2372) - (-2.6933)} = 2.7072$$

$$\text{Error} = |x_k - x_{k-1}| = |2.7072 - 2.6933| = 0.0139$$

Here, $\epsilon > \text{Error}$, therefore we perform fourth iteration

Putting the values in a tabular form we get

k	x_{i-1}	$f(x_{i-1})$	x_i	$f(x_i)$	x_{i+1}	$f(x_{i+1})$	$ x_k - x_{k-1} $
1	2	-9	3	6	2.6	-1.824	-
2	3	6	2.6	-1.824	2.6933	-0.2372	0.0933
3	2.6	-1.824	2.6933	-0.2372	2.7072	0.012	0.0139

Iteration 4:

$$k = 4, x_{i-1} = 2.6933, x_i = 2.7072,$$

$$f(x_{i-1}) = f(2.6933) = (2.6933)^3 - 4(2.6933) - 9 = -0.2372$$

and

$$f(x_i) = f(2.7072) = (2.7072)^3 - 4(2.7072) - 9 = -0.2372$$

Fourth approximation to the root is

$$x_{i+1} = \frac{x_{i-1}f(x_i) - x_i f(x_{i-1})}{f(x_i) - f(x_{i-1})} = \frac{(2.6933)(0.012) - (2.7072)(-0.2372)}{0.012 - (-0.2372)} = 2.7065$$

$$\text{Error} = |x_k - x_{k-1}| = |2.7065 - 2.7072| = 0.0007$$

Here, $\text{Error} < \epsilon$

Hence, the root is 2.7065, correct to two decimal places.

Putting the values in a tabular form we get

k	x_{i-1}	$f(x_{i-1})$	x_i	$f(x_i)$	x_{i+1}	$f(x_{i+1})$	$ x_k - x_{k-1} $
1	2	-9	3	6	2.6	-1.824	-
2	3	6	2.6	-1.824	2.6933	-0.2372	0.0933
3	2.6	-1.824	2.6933	-0.2372	2.7072	0.012	0.0139
4	2.6933	-0.2372	2.7072	0.012	2.7065	0	0.0007

Example

Find the real root of the equation $x^3 - x - 1 = 0$ by Secant Method correct. Take $x_0 = 1, x_1 = 1.5, \epsilon = 0.001$

k	x_{i-1}	$f(x_{i-1})$	x_i	$f(x_i)$	x_{i+1}	$f(x_{i+1})$	$ x_k - x_{k-1} $
1	1	-1	1.5	0.875	1.2667	-0.2344	-
2	1.5	0.875	1.2667	-0.2344	1.316	-0.037	0.0493
3	1.2667	-0.2344	1.316	-0.037	1.3252	0.0021	0.0092
4	1.316	-0.037	1.3252	0.0021	1.3247	0	0.0005

$$\epsilon = 0.001, |x_k - x_{k-1}| = 0.0005$$

Hence, the root is 1.3247, correct to three decimal places.

Example

This polynomial obviously has roots at $x = 2$ and at $x = 4$; one is a double root, the other a triple root:

$$\begin{aligned} f(x) &= (x-2)^3(x-4)^2 \\ &= x^5 - 14x^4 + 76x^3 - 200x^2 + 256x - 128 \end{aligned}$$

- Which root can you get with bisection? Which root can't you get?
- Repeat part (a) with the secant method. Ans. Both roots can be reached with secant method
- If you begin with the interval $[1, 5]$, which root will you get with
 - bisection,
 - the secant method,
 - false position?

d. Use Newton's method with $x_0 = 3$. Does it converge? To which root?

Solution

- we can get the root 2. Getting Interval for 4 is not possible.
- Both roots can be reached with secant method
- (1) Converge to 2, (2) It will converge to 2 (3) Converge to 2
- Derivate is zero, so the method does not converge. Yes $x_0 = 4.1$, can converge to $x = 4$

x_i	$f(x_i)$	$f'(x_i)$	x_{i+1}	$ c_k - c_{k-1} $
4.1	0.0926	1.9845	4.0533	
4.0533	0.0246	0.9594	4.0277	0.0256
4.0277	0.0064	0.4707	4.0141	0.0136
4.0141	0.0016	0.233	4.0071	0.007
4.0071	0.0004	0.1159	4.0036	0.0035

Questions

1. The smallest positive root of the equation

- $f(x) = x^4 - 3x^2 + x - 10 = 0$
 - Find an interval of unit length which contains this root.
- Find the root using any of the methods

2. Find the root of the following using Bisection Method:

- $f(x) = x^3 - x - 1 = 0$
- $f(x) = xe^x = 1$

3. Find the root of the following using Newton Raphson Method:

- Find an approximation to $\sqrt{5}$ to four decimal places

4. Find the root of using Bisection method, False Position, Newton-Raphson, Secant upto accuracy 0.001.

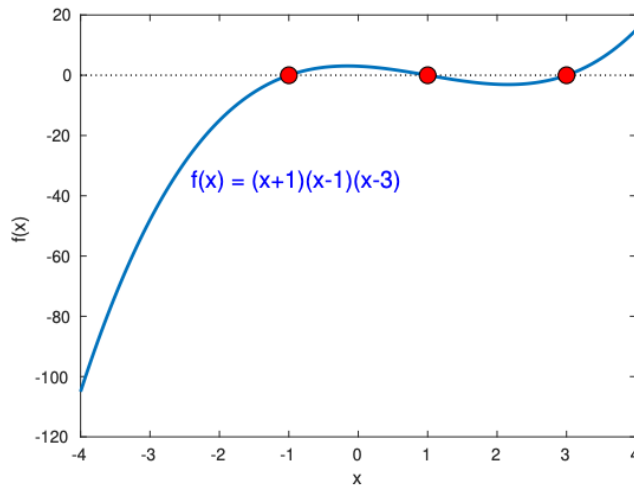
5. $x^3 - x - 4 = 0$

Ans. 1.7963219

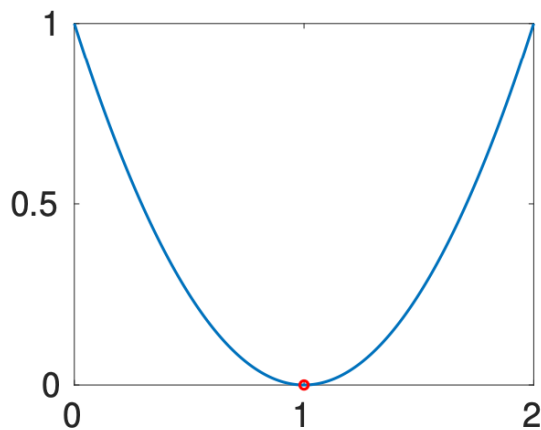
6. $x^4 - x - 10 = 0$

Ans. 1.8556

7. The function $f(x) = (x+1)(x-1)(x-3)$ is pictured in the plot. If the bisection algorithm is applied with initial interval $[-4, 4]$, Which root of $f(x)$ will the bisection method converge to? Which intervals can you take to find the other roots? Find the roots using bisection method.



8. Let $f(x) = x^2 - 2x + 1$. Can bisection method be used to approximate the root of the function $f(x)$ pictured.



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